

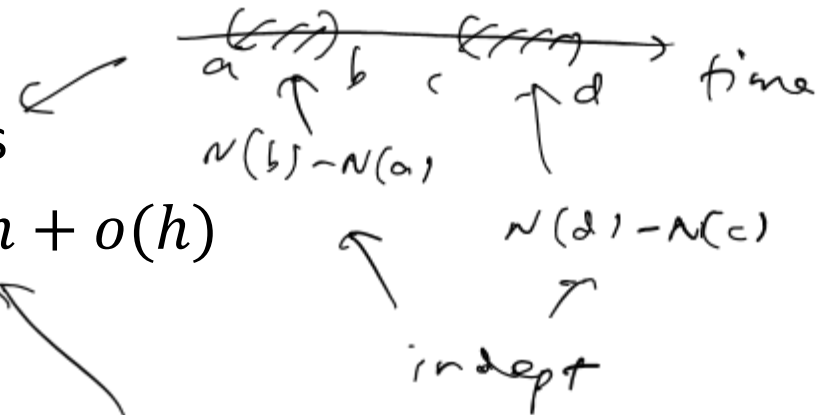
IEOR 4404: Simulation

Lecture 12: Simulating Time Inhomogeneous
Poisson Processes and Discrete-Event
Simulation

Time Inhomogeneous Poisson Process

A counting process with the following properties:

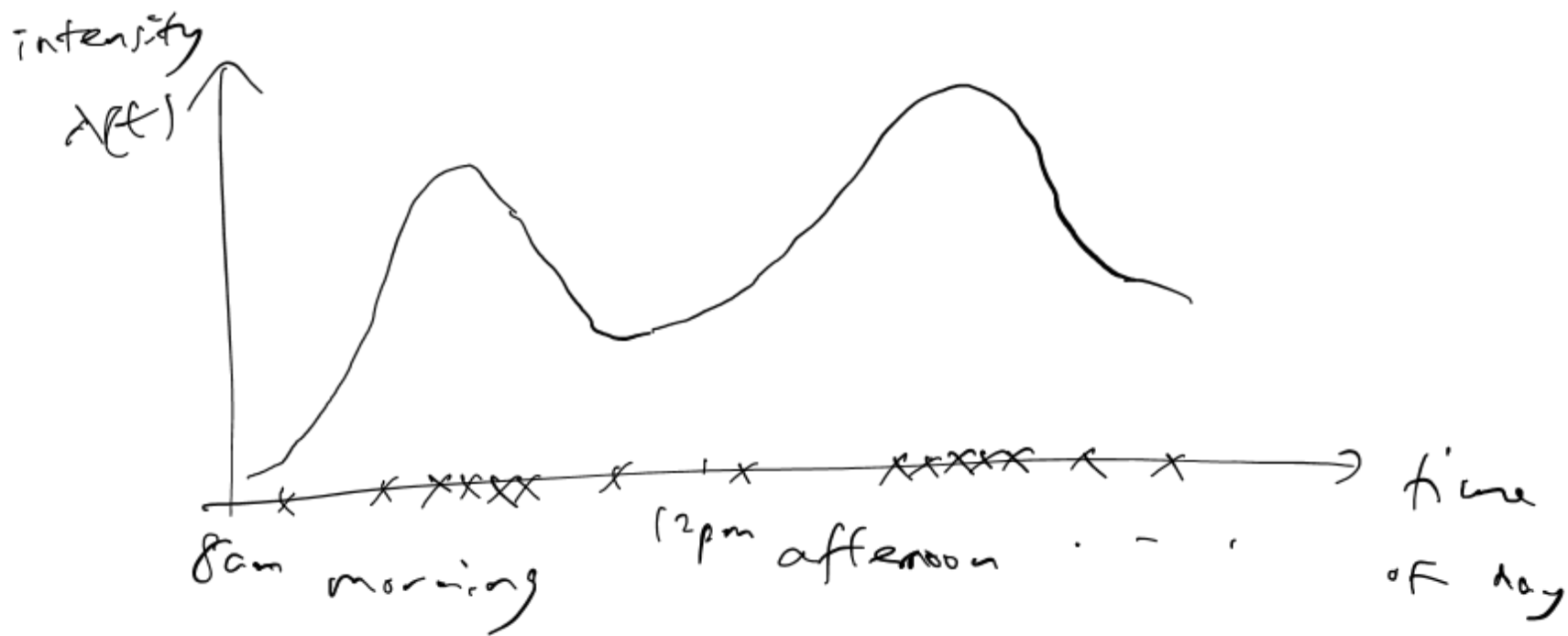
- $N(0) = 0$
- $N(t)$ has independent increments
- $P(N(t+h) - N(t) = 1) = \lambda(t)h + o(h)$
- $P(N(t+h) - N(t) \geq 2) = o(h)$



The function $\lambda(t)$ is called the intensity

in a regular Poisson process,
this is just λ .

Compared to standard homogeneous Poisson process, inhomogeneous Poisson process has independent increments but not necessarily stationary increments



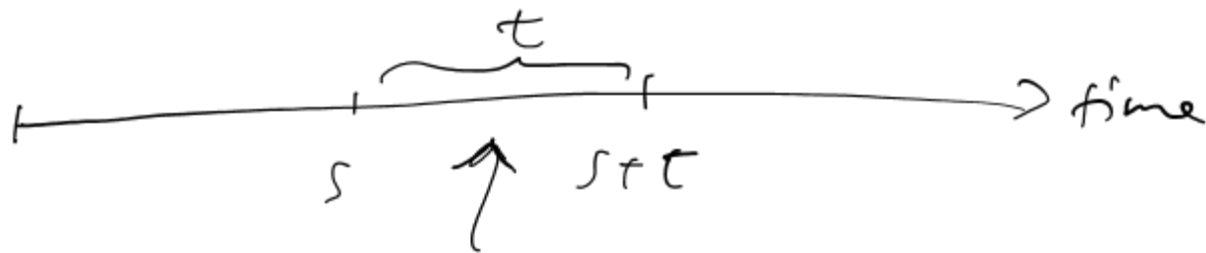
Time inhomogeneous Poisson Process

For an inhomogeneous Poisson process with intensity $\lambda(t)$, its increment over $(s, s + t]$, i.e., $N(s + t) - N(s)$, is distributed according to $Poi(m(s + t) - m(s))$ where

$$m(t) = \int_0^t \lambda(y) dy$$

is the mean value function

This is from a generalization of the Poisson approximation to binomial or the “law of small numbers”



$$\# \text{ arrivals} = N(t+s) - N(s)$$

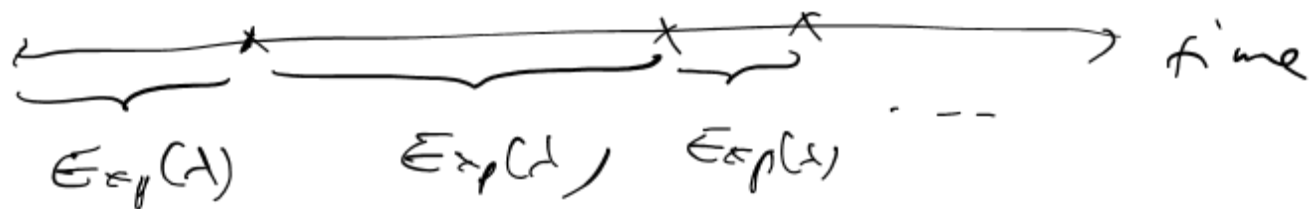
$$N(t+s) - N(s) \sim \text{Poi}\left(\int_s^{t+s} \lambda(u) du\right)$$

For regular Poisson process with rate λ , set $\lambda(t) = \lambda$.

$$\Rightarrow \int_s^{t+s} \lambda(u) du = \int_s^{t+s} \lambda du = \lambda t$$

For regular Poisson process,

- ① use interarrival times:
they are iid $\text{Exp}(\lambda)$



- ② Generate $N(T) \sim \text{Poi}(\lambda T)$ and place the arrival times as order statistic of $\text{Unif}(0, T)$

"Proposal" approach:

Modify $\text{Exp}(\lambda)$ to $\text{Exp}(\lambda(\epsilon)) \dots$

This fails...

Simulating Time Inhomogeneous Poisson Process

Recall the two approaches to simulate standard Poisson processes:

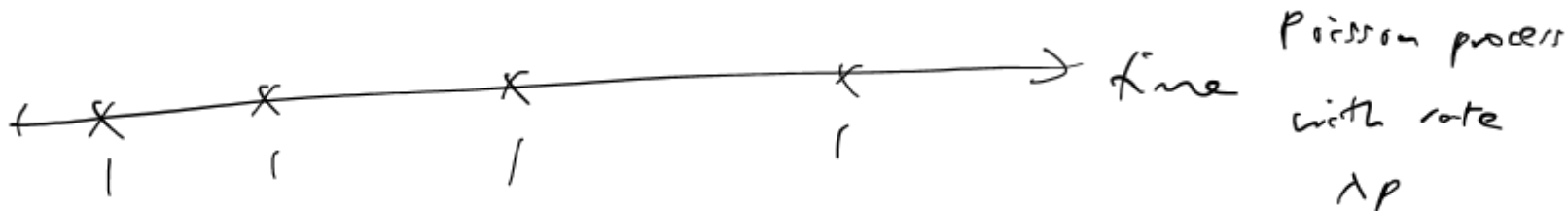
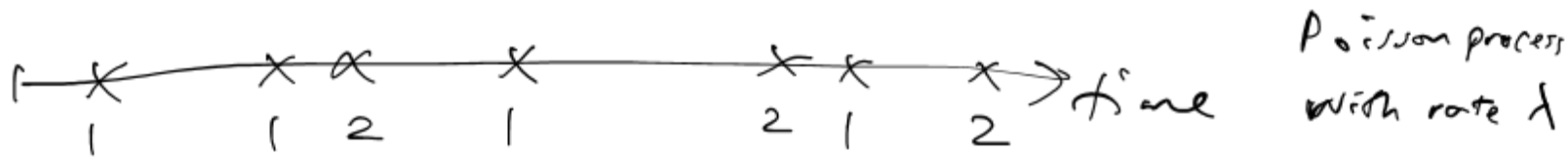
1. Generate the interarrival times via exponential variables:
 - Generate $X_i \sim \text{Exp}(\lambda), i = 1, \dots, n + 1$ until we see $X_1 + \dots + X_n \leq T, X_1 + \dots + X_n + X_{n+1} > T$
 - Then $A_i = X_1 + \dots + X_i, i = 1, \dots, n$
2. Use conditional representation of arrival times as order statistics:
 - Generate $N(T) \sim \text{Poi}(\lambda T)$
 - Suppose $N(T) = n$. Generate n i.i.d. $\text{Unif}(0, T)$ and sort to get $U_{(1)} < \dots < U_{(n)}$

Can we use these approaches to simulate inhomogeneous Poisson processes?

Thinning

For a standard Poisson process with rate λ , suppose each event is of Type 1 or 2 with probability p and $1 - p$ respectively, independently of all other dynamics.

Then the counting process of Type 1 events and the counting process of Type 2 events are Poisson processes with rate λp and $\lambda(1 - p)$ respectively. Moreover, these two processes are independent.

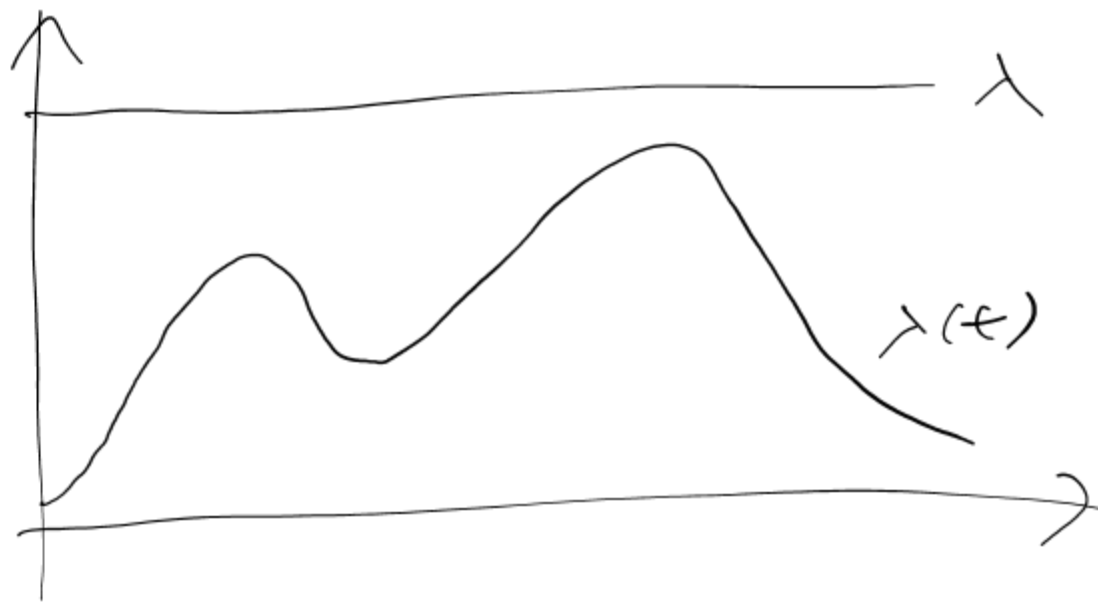


Thinning

This notion can be generalized substantially:

If we have m types of events, each with probability $p_i, i = 1, \dots, m$. Then the counting process of each type of events is a Poisson process with rate λp_i . Moreover, all these processes are independent.

Consider the case where an event in a standard Poisson process is marked as a “success” with probability $P(t)$ at time t . Then the counting process of successful events follows an inhomogeneous Poisson process with intensity $\lambda P(t)$



- Simulate Poisson process with rate λ
- For each arrival, label "success" with probability $\frac{\lambda(t)}{\lambda}$
- Then output is time-inhomogeneous Poisson process with intensity $\lambda \cdot \frac{\lambda(t)}{\lambda} = \lambda(t)$

Simulation via Thinning

The last thinning property connects inhomogeneous Poisson process with standard Poisson process

To simulate an inhomogeneous Poisson process, do the following:

- Find λ such that $\lambda(t) \leq \lambda$
- Simulate a standard Poisson process with rate λ
- For each incurred event, accept with probability $P(t) = \frac{\lambda(t)}{\lambda}$ where t is the event time

Simulating Time Inhomogeneous Poisson Process

Applying thinning to Approach 1 for simulating standard Poisson process:
Simulate interarrival and hence arrival times via exponential variables,
and “accept” or “reject” them

Find λ such that $\lambda(t) \leq \lambda$ for all $t \leq T$

use interarrival times

Set $t = 0, I = 0$

While $t \leq T$,

- Generate $U \sim \text{Unif}(0,1)$
 - $t = t - \frac{1}{\lambda} \log U$
 - Generate $V \sim \text{Unif}(0,1)$
 - If $V \leq \frac{\lambda(t)}{\lambda}$, then $I = I + 1, A(I) = t$
- } simulate standard Poisson process
- } doing “acceptance-rejection”

Output $I - 1$ and $A(i), i = 1, \dots, I - 1$

Simulating Time Inhomogeneous Poisson Process

Applying thinning to Approach 2 for simulating standard Poisson process:
Simulate total number of arrivals, place them uniformly and “accept” or “reject” them

Find λ such that $\lambda(t) \leq \lambda$ for all $t \leq T$

use conditional
representation

Initialize $\mathcal{A}$ as an empty set

- Generate $N(T) \sim \text{Poi}(\lambda T)$
- Suppose $N(T) = n$. Generate n i.i.d. $\text{Unif}(0,1)$, say $U_1, \dots, U_n$
- Sort $U_1, \dots, U_n$ in increasing order to get $U_{(1)} < \dots < U_{(n)}$
- Set $A_i' = TU_{(i)}, i = 1, \dots, n$
- Generate n i.i.d. $V_i \sim \text{Unif}(0,1)$
- For each $i = 1, \dots, n$, if $V_i \leq \frac{\lambda(A_i')}{\lambda}$, then put A_i' into $\mathcal{A}$
- Output $\mathcal{A}$ as the set of event times

arrival times for the “virtual” standard Poisson process

Simulating Time Inhomogeneous Poisson Process

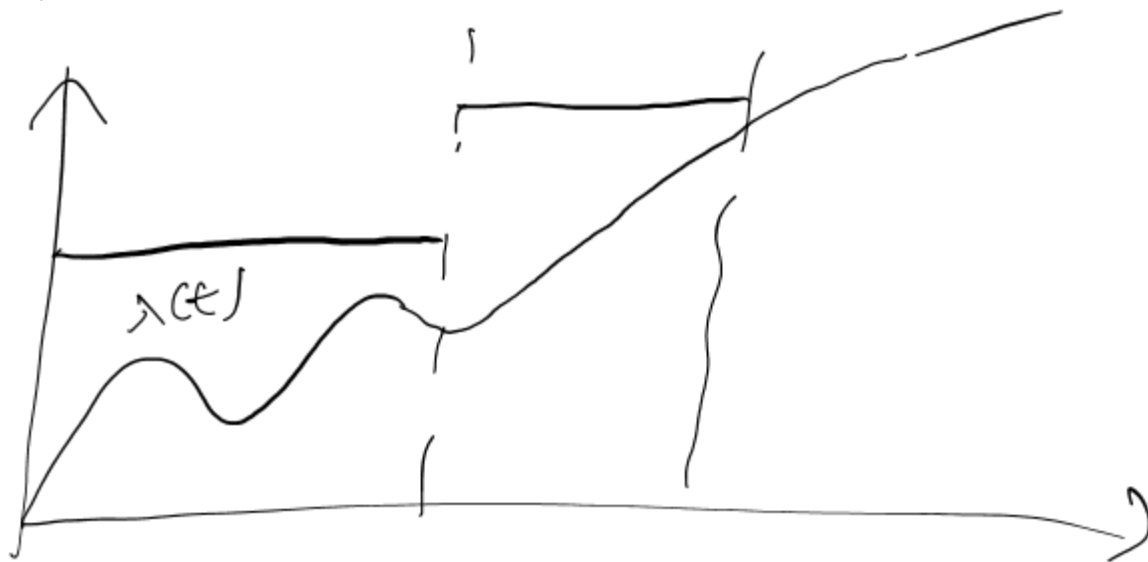
Modification of the second approach: Simulate total number of arrivals, and place them according to a distribution proportional to $\lambda(t)$

- Generate $N(T) \sim Poi(\int_0^T \lambda(t) dt)$
- Suppose $N(T) = n$. Generate n i.i.d. variables, say $V_1, \dots, V_n$, from a density proportional to $\lambda(t)$, or more precisely

$$\frac{\lambda(t)}{\int_0^T \lambda(t) dt}, \quad \text{for } 0 \leq t \leq T$$

- Sort $V_1, \dots, V_n$ in increasing order to get $V_{(1)} < \dots < V_{(n)}$
- These are the event times

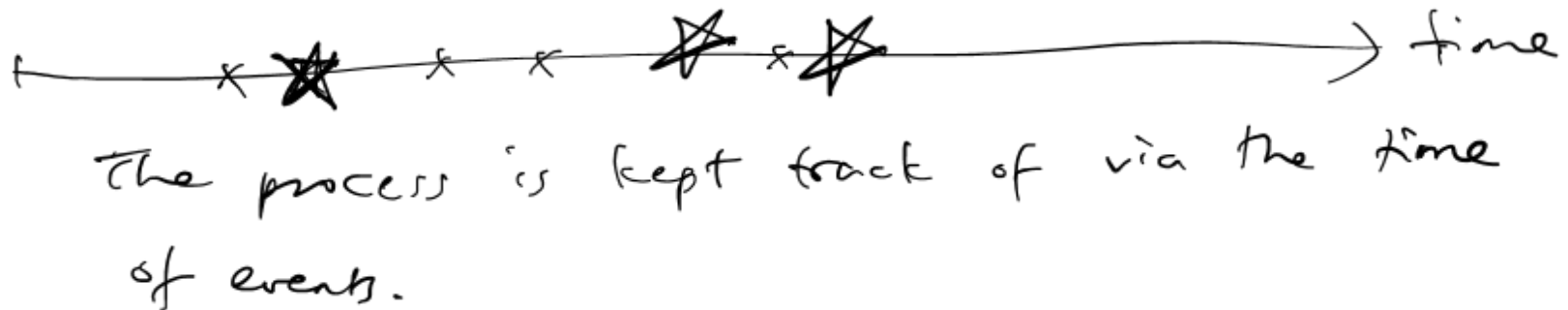
Suppose $\lambda(t)$ is not bounded.



Divide the time into intervals, and within each interval there is a bound λ . Then we can simulate each interval separately.

Discrete-Event Simulation

- We can simulate processes that are driven by events happening at different times
- These processes can be encoded by the event types and times
- Examples include M/M/1 and M/M/s queues
- For systems that are beyond Markov chains or any “nice” stochastic processes, we can still simulate them (in fact, this is what makes simulation powerful)



Discrete-Event Simulation

First-come-first-serve G/G/1 queue: Interarrival times and service times each follow own distributions independently

To simulate the waiting time of customers, we use the recursion

$$W_{n+1} = (W_n + S_n - X_{n+1})^+ \text{ for } n \geq 0$$

where

- W_n = waiting time of customer n
- S_n = service time of customer n
- X_{n+1} = interarrival time between customer n and $n + 1$

and we set

- $W_0 = 0, S_0 = 0$ and X_1 = first arrival time

*This process is not
discrete state-space
MC, so it's beyond
the classical MC analysis.*

Discrete-Event Simulation

This is called the Lindley's recursion

For more complex queueing systems, similar type of recursion holds (but more complicated)

Many other quantities, such as queue length, departure times etc., can be obtained from the simulated information

We may be interested in:

- Transient quantities, e.g., expected waiting time of the n -th customer: Repeat many simulation replications and take average (and output confidence intervals)
- Steady-state quantities, e.g., long-run expected waiting time. Simulate a long trajectory and take average after removing burn-in